SIM2001 Advanced Calculus : Tutorial 6

1. Find the spherical coordinate limits for the integral that calculates the volume of the given solid and then evaluate the integral. [See Textbook for the figures]

   (14.7.55) The solid between the sphere $\rho = \cos\phi$ and the hemisphere $\rho = 2$, $z \geq 0$

   (14.7.57) The solid enclosed by the cardioid of revolution $\rho = 1 - \cos\phi$

   (14.7.59) The solid bounded below by the sphere $\rho = 2\cos\phi$ and above by the cone $z = \sqrt{x^2 + y^2}$

2. (14.7.71)
   Find the volume of the portion of the solid sphere $\rho \leq a$ that lies between the cones $\phi = \dfrac{\pi}{3}$ and $\phi = \dfrac{2\pi}{3}$.

3. (14.7.73)
   Find the volume of the smaller region cut from the solid sphere $\rho \leq 2$ by the plane $z = 1$.

4. (14.7.75)
   Find the volume of the region bounded below by the plane $z = 0$, laterally by the cylinder $x^2 + y^2 = 1$, and above by the paraboloid $z = x^2 + y^2$.

5. (14.7.81)
   Find the volume of the region bounded above by the paraboloid $z = 5 - x^2 - y^2$, below by the paraboloid $z = 4x^2 + 4y^2$.

6. (14.7.83)
   Find the volume of the region cut from the solid cylinder $x^2 + y^2 \leq 1$ by the sphere $x^2 + y^2 + z^2 = 4$.

7. (14.8.1)

   (a) Solve the system $u = x - y$, $v = 2x + y$ for x and y in terms of u and v. Then Find the value of the Jacobian $\dfrac{\partial(x, y)}{\partial(u, v)}$.

   (b) Find the image under the transformation $u = x - y$, $v = 2x + y$ of the triangular region with vertices $(0, 0)$, $(1, 1)$, and $(1, -2)$ in the xy-plane. Sketch the transformed region in the uv-plane.

8. (14.8.3)

   (a) Solve the system $u = 3x + 2y$, $v = x + 4y$ for x and y in terms of u and v. Then Find the value of the Jacobian $\dfrac{\partial(x, y)}{\partial(u, v)}$.

   (b) Find the image under the transformation $u = 3x + 2y$, $v = x + 4y$ of the triangular region in the xy-plane bounded by the x-axis, the y-axis and the line $x + y = 1$. Sketch the transformed region in the uv-plane.

9. (14.8.7)
   Use the transformation $u = 3x + 2y$, $v = x + 4y$ to evaluate the integral

$$\iint_R (3x^2 + 14xy + 8y^2)dxdy$$

   for the region R in the first quadrant bounded by the lines

$$y = -\frac{3}{2}x + 1, \quad y = -\frac{3}{2}x + 3, \quad y = -\frac{1}{4}x, \quad y = -\frac{1}{4}x + 1.$$

10. (14.8.9)
Let R be the region in the first quardrant of the xy-plane bounded by the hyperbolas $xy = 1$, $xy = 9$, and the lines $y = x$, $y = 4x$. Use the transformation $x = \dfrac{u}{v}$, $y = uv$ with $u > 0$ and $v > 0$ to rewrite

$$\iint_R \left(\sqrt{\frac{y}{x}} + \sqrt{xy}\right) dxdy$$

as an integral over an appropriate region G in the uv-plane. Then evaluate the uv-integral over G.

11. (14.8.13)
Use the transformation $u = x + 2y$, $v = x - y$ to evaluate the integral

$$\int_0^{2/3} \int_y^{2-2y} (x + 2y)e^{y-x} dxdy$$

by first writing it as an integral over a region G in the uv-plane.

12. (14.8.19)
Evaluate $\iiint |xyz| dxdydz$ over the solid ellipsoid $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} + \dfrac{z^2}{c^2} \leq 1$.
(*Hint:* Let $x = au$, $y = bv$ and $z = cw$. Then integrate over an appropriate region in uvw-space.)